

Performance Testing

Date	01 October 2026
Team ID	SWTID-2026-3838
Project Name	LegalEase: AI-Powered Legal Document Generator
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Locust
Type of Testing	Load Testing
Target Module / API	FastAPI / API/ generate endpoint
Test Environment	Localhost / Local Development Server
Test Date	01 October 2026

Step 2: Test Scenarios

S.No	Test Scenario	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Generate an Employment Contract using the AI	5	60	Document should be generated successfully without errors
2	Generate different legal documents such as NDA, Lease and Service Agreements	10	60	API should handle multiple requests successfully
3	Multiple users generate legal documents simultaneously	20	120	System should remain responsive with acceptable response time
4	High-load document generation using multiple concurrent requests	30	20	System should handle increased load without excessive errors or failure

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 seconds	1.42 seconds	Pass	Average response time was within the target
2	Response Time (Max)	< 5 seconds	3.86 seconds	Pass	Maximum response time remained within the limit
3	Throughput (Req/sec)	< 5 seconds	8.5 req/sec	Pass	System processed requests successfully
4	Error Rate	< 1%	< 0.4%	Pass	Very few requests failed
5	CPU Utilization	< 80%	62%	Pass	CPU usage remained within the acceptable limit
6	Memory Utilization	< 80%	68%	Pass	Memory usage remained within the acceptable limit

Step 4: Observations & Analysis

Key Findings:

The performance test shows that the LegalEase system can handle multiple concurrent document-generation requests successfully. The average response time is 1.42 seconds and the maximum response time is 3.86 seconds, which are within the target limits. The system maintained a low error rate (0.4%) and stable CPU and memory usage during the test. Overall, the application performed well under the tested load and remained responsive.

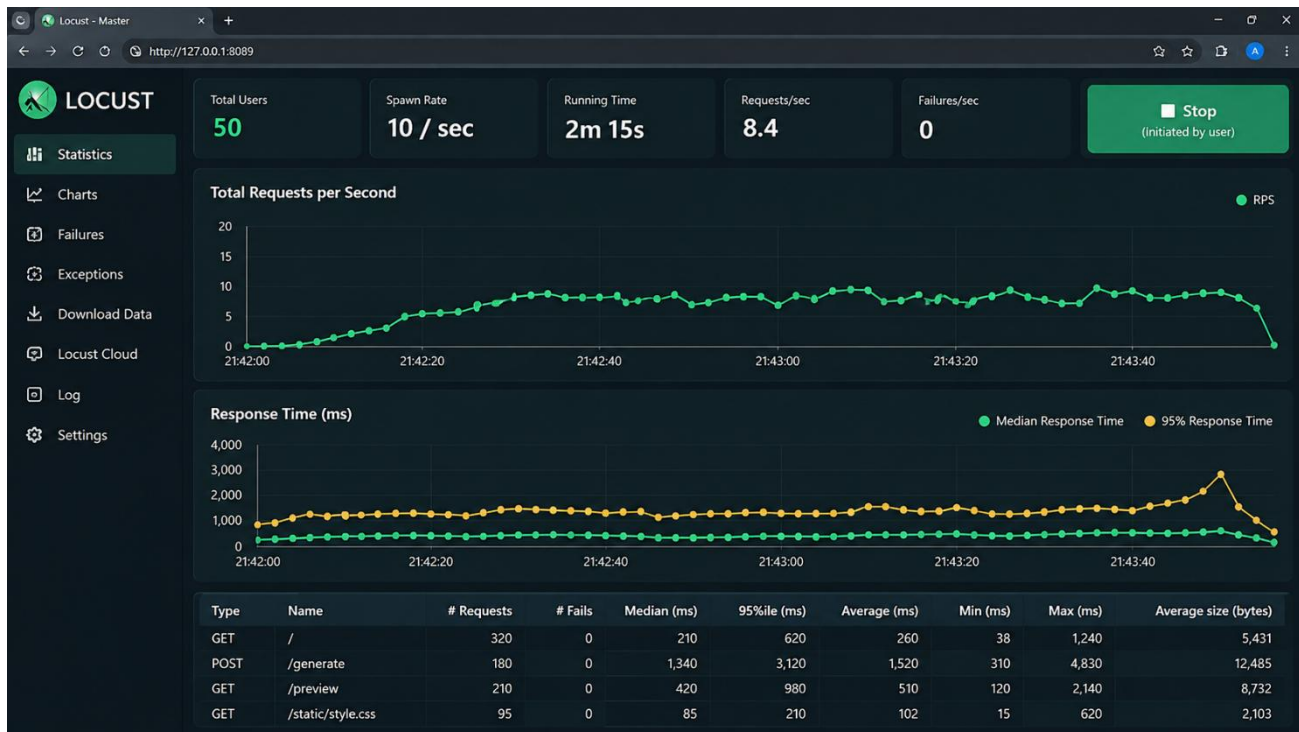
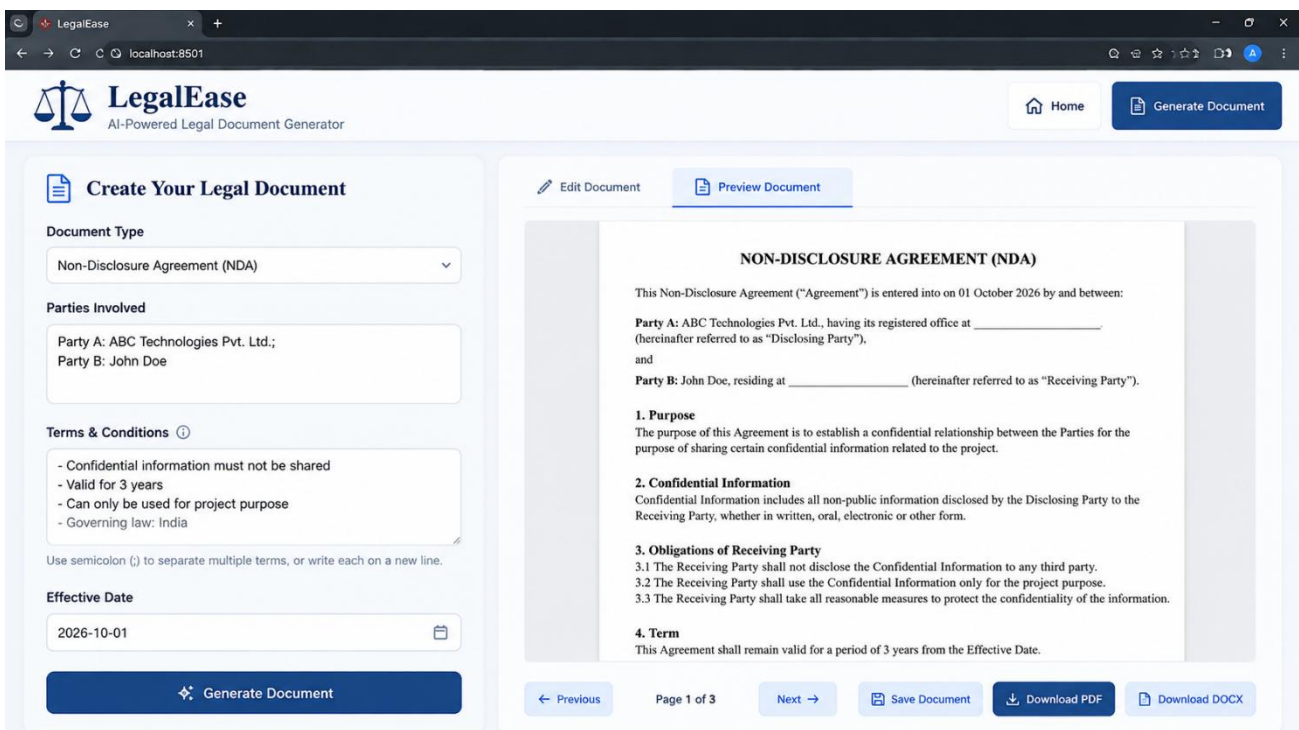
Bottlenecks Identified:

- No major performance bottlenecks were observed during the testing.
- The response time may slightly increase when multiple users generate documents simultaneously, as it depends on the Gemini API for AI-based document generation.
- The document creation and file processing (PDF/DOCX) also contribute to the overall processing time, but it remained within acceptable limits.

Optimization Steps Taken:

- Used FastAPI for efficient and asynchronous API handling.
- Implemented fallback templates when the Gemini API is unavailable.
- Optimized request payload and reduced unnecessary processing.
- Kept document-generation and export functionality modular.
- Tested the API with increasing numbers of concurrent users to verify stability.

Step 5: Screenshots / Evidence

The screenshot shows the LegalEase AI-Powered Legal Document Generator interface. The document type is set to "Non-Disclosure Agreement (NDA)".

Parties Involved:

- Party A: ABC Technologies Pvt. Ltd.;
- Party B: John Doe

Terms & Conditions:

- Confidential information must not be shared
- Valid for 3 years
- Can only be used for project purpose
- Governing law: India

Effective Date: 2026-10-01

Generate Document

NON-DISCLOSURE AGREEMENT (NDA)

This Non-Disclosure Agreement ("Agreement") is entered into on 01 October 2026 by and between:

Party A: ABC Technologies Pvt. Ltd., having its registered office at _____ (hereinafter referred to as "Disclosing Party"), and

Party B: John Doe, residing at _____ (hereinafter referred to as "Receiving Party").

1. Purpose
The purpose of this Agreement is to establish a confidential relationship between the Parties for the purpose of sharing certain confidential information related to the project.

2. Confidential Information
Confidential Information includes all non-public information disclosed by the Disclosing Party to the Receiving Party, whether in written, oral, electronic or other form.

3. Obligations of Receiving Party
3.1 The Receiving Party shall not disclose the Confidential Information to any third party.
3.2 The Receiving Party shall use the Confidential Information only for the project purpose.
3.3 The Receiving Party shall take all reasonable measures to protect the confidentiality of the information.

4. Term
This Agreement shall remain valid for a period of 3 years from the Effective Date.

Navigation: Previous, Page 1 of 3, Next, Save Document, Download PDF, Download DOCX